

The Road in Front of the Ship

A toy metric profile that “shortens” local space 1000× — and the engine that refuses to believe it

because writing the metric down is the easy part; paying for it is the point

A toy-geometry and screening note — written to be checked by a skeptical relativist, and read by everyone else

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What this note is — and is not. This is a **toy metric profile and a screening story** — not a working drive, not a build method, not faster-than-light travel, not a hardware recipe, not an energy source, and not a Mars travel-time result. Writing down a metric that locally shortens proper distance is mathematically trivial, and doing so is **not** a discovery. The interesting question is not “can we build this?” but “what would *source* it, and what gate kills it?” The ceiling anywhere in this note is a toy result; the strongest verb is *screen*.

Before we start

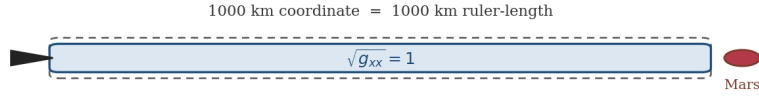
The other notes in this series checked whether the geometry reproduces something real — light bending, hydrogen, the forces. This last one is different, and more fun, and it comes with a sharper warning. It takes the most seductive idea in speculative propulsion — bend space so the road ahead is shorter — writes down the simplest toy that expresses it, and then does the one thing that separates honest work from science fiction: it asks who pays for the geometry, and lets the answer kill the idea.

The hero here is not a ship. It is a screening engine — the constraint-first ‘First-Failure Engine’ behind the attached provisional patent — whose entire job is to refuse to believe its own most tempting result until every gate allows it. You will trust it not because it gives you a drive (it does not), but because, at the one moment it could have sold you a result, it instead names the single number it cannot pay for and kills its own best candidate. That refusal is the point. You can read every word skipping every equation.

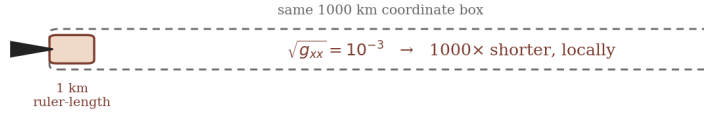
1. Space as a road with variable ruler-density

Start with what ‘distance’ means. In ordinary geometry, distance is not just how you label coordinates — it is what the metric says a ruler measures. On a flat road, one coordinate kilometre is one kilometre of ruler-length: $d\ell = dx$, and a coordinate distance D has proper length $L = D$.

ordinary space



toy front-lobe profile



distance is what the metric says a ruler measures — the toy shrinks the ruler-length per coordinate metre in the forward lobe **TOY**

Figure 1. *The toy, in one picture. Top: in ordinary space a coordinate kilometre is a kilometre of ruler-length. Bottom: the toy front-lobe profile keeps the same coordinate span but sets $\sqrt{g_{xx}} = 10^{-3}$, so the lobe holds only a thousandth of the ruler-length — a $1000\times$ local shortening. This is a toy, not a claim.*

Now the toy. Suppose, in a patch of road directly ahead of the ship, the metric were $d\ell^2 = \lambda^2 dx^2$ with $\lambda = 1/1000$. Then $d\ell = dx/1000$, and that patch’s proper length is $L/1000$ — a thousand coordinate kilometres measuring just one kilometre of ruler-length. Not because the map was redrawn or the coordinate relabelled, but because the metric says rulers measure less length through that region. That is the entire toy idea: change the local spatial metric in front of the vehicle so the proper-length density is smaller.

2. The careful version of “space density”

It is tempting to call this ‘changing the density of space,’ but that phrase invites a wrong picture — it is not mass density. The precise statement uses the proper-length density $\rho_L(x) \equiv \sqrt{g_{xx}}$, so that $d\ell = \rho_L dx$. Ordinary space has $\rho_L = 1$ — one metre of ruler-length per coordinate metre. The toy front lobe has $\rho_L = 10^{-3}$ — one millimetre of ruler-length per coordinate metre. That is the cleanest honest version of ‘space density’: not the physical density of space, but how much ruler-length a coordinate interval carries.

3. The front lobe, not the whole highway

The picture is deliberately local. The ship does not drag all of Earth–Mars space into a new metric; it carries — or tracks — a compact metric-interface region immediately ahead, and only the forward lobe is altered. In vehicle-comoving coordinates $\xi = x - x_{\text{ship}}(t)$, the lobe lives in a small window ahead of the ship, with a smooth compression profile $\lambda(\xi)$ dipping toward 10^{-3} inside the lobe and returning to 1 outside.

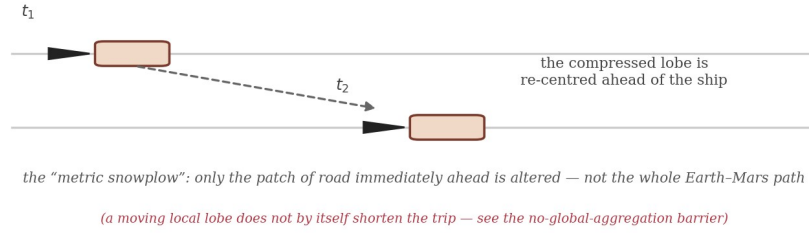


Figure 2. The “metric snowplow.” Only the patch of road immediately ahead is altered, re-centred as the ship moves — not the whole Earth-Mars path. The caveat is built in: a moving local lobe does not by itself shorten the trip (Section 7).

Call it the metric snowplow: not folding the whole highway, not teleporting Mars, not shortening the universe — just changing the ruler-density in the patch of road immediately ahead, continuously re-centred as the ship moves. Whether a moving local lobe actually buys a shorter trip is a separate, much harder question — Section 7 — and the honest answer, so far, is no.

4. Route one: the ordinary-GR toy (the easy part)

Here is the toy computed the ordinary way, with no geometry beyond freshman calculus. The proper length through an interval is $L = \int \sqrt{g_{xx}} dx$. With $g_{xx} = 1$, $L = b - a$. With $g_{xx} = 10^{-6}$ across the lobe, $\sqrt{g_{xx}} = 10^{-3}$, so $L = 10^{-3}(b - a)$ — a factor of a thousand shorter. The local shortening score is $S_L = 1 - 10^{-3} = 0.999$, a 99.9% local proper-distance deficit.

And here is the first honest beat: that was trivial. Writing down a metric that locally shortens proper distance is an undergraduate exercise. As the provisional patent puts it, it is trivially easy to write down a spacetime metric and assert that it produces a desirable kinematic effect — while quietly skipping the part that matters. The whole value of what follows is in **not** skipping that part.

5. Route two: the same target from the 13D platform

The series’ usual move — compute the same thing from the geometry — works here too, but with a twist. The 13D platform does not begin by saying ‘build this.’ It treats the toy as a candidate data object Ξ_{drive} (a bundle of metric and field perturbations), projects it into the ordinary 4D metric $g_{\mu\nu} = g_{bg} + \delta g$, and evaluates the same forward proper-distance functional $D[\Xi] = \int \sqrt{g_{\xi\xi}} d\xi$. For the front-lobe toy this is the same integral as before, and the two routes meet at the same local geometry: $D[\Xi]_{\text{compressed}} = 10^{-3} D[\Xi]_{\text{normal}}$.

But meeting on the geometry is meeting on the easy part. The 13D route’s actual job begins exactly where the ordinary toy stops — by refusing to call a written-down metric a result.

6. The killer question: who pays for the metric?

Once the geometry is fixed, the stress-energy it demands is not optional. Einstein's equation runs both ways: $G_{\mu\nu} = 8\pi G T_{\mu\nu}$ means the geometry on the left forces a source on the right, $T_{\text{req}} = G[g]/8\pi G$. So the engine reconstructs the stress-energy this front-lobe geometry would demand and asks: can any declared admissible field supply it? Does it violate the energy conditions? Does it need negative energy or an undeclared current?

This is where the engine earns its name — and it has already run the experiment on a smaller, honestly worked toy called MID-METRIC-001. That candidate is not a strawman: it produces a real, invariant, sub-luminal forward-distance deficit of about 0.22%, with nonzero curvature ($R \neq 0$) proving it is no coordinate mirage — every check that would expose a fake passes. A weaker document would headline exactly this.

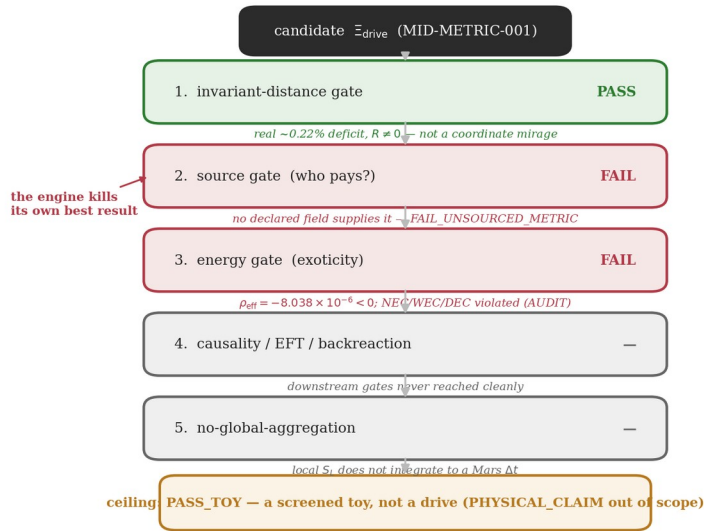


Figure 3. The First-Failure Engine, on its own best candidate. MID-METRIC-001 passes the invariant-distance gate (a real local deficit, not a coordinate mirage) and then dies at the source gate — no declared field supplies the demanded stress-energy — with exotic negative energy as the root cause at the energy gate. The ceiling stays PASS_TOY; a physical claim is out of scope.

Then the engine asks who pays, and the answer is a single number it cannot source: a negative effective energy density, $\rho_{\text{eff}} = -8.038 \times 10^{-6}$ at the lobe centre, with the null, weak, and dominant energy conditions all violated, supplied by no declared field. The geometry gate passes; the source gate fails — terminal token FAIL_UNSOURCED_METRIC — with the exotic negative energy as the root cause at the energy gate (status AUDIT, never a clean pass).

The engine kills its own best result. That refusal — turning a seductive effect into a named failure by one computed number, rather than a sales line — is the entire

point of the exercise. The fun $1000\times$ profile is not a claim of success; it is a target for the engine to attack, and the engine wins.

7. The other trap: local is not global

There is a second way to overclaim, subtler than the source gate, and the engine blocks it by theorem. Even granting a local shortening, a moving lobe does not automatically buy a shorter trip to Mars. A positive local shortening score S_L does not integrate into a global travel-time reduction Δt — the corpus calls this the no-global-aggregation result — and any Earth-Mars travel-time number built directly from local S_L is flagged `FAIL_MISSION_OVERCLAIM`. So the honest reading refuses two tempting leaps at once: from ‘I can write the metric’ to ‘I can source it’ (the source gate says no), and from ‘the lobe is locally shorter’ to ‘the trip is shorter’ (the aggregation barrier says not without much more).

8. What this does, and does not, show

What it shows: a $1000\times$ local distance-shortening target is easy to express — set $\sqrt{g_{xx}}$ to 10^{-3} in the forward lobe — and the 13D platform can represent it as a candidate and compute the same local deficit as ordinary GR.

What it does not show: that the profile is sourceable, that energy conditions hold, that the lobe can be generated, that a moving lobe shortens a Mars trip, or — emphatically — a drive. It provides no hardware, no energy source, no build instructions; those are out of scope by construction, not by modesty.

The honest status is a target-profile and screening story. The ceiling is a toy result (`PASS_TOY`); the top rung that would mean a physical claim is explicitly out of scope and targeted nowhere. The strongest verb is screen.

9. The seam: the engine is the product, not the road-shortening trick

So the hard problem was never writing down the geometry — that is the undergraduate part. The hard problem is paying for it, and the toy does not. What survives is not a drive but a discriminator: a fail-closed engine that takes a seductive metric, reconstructs the source it would demand, audits it against admissible fields and the energy conditions, and refuses to promote it past a toy label — an engine whose every incompleteness can only downgrade a candidate, never promote one.

That engine, not the road-shortening trick, is the thing of value, and it is the subject of the attached provisional patent: a screening and source-ledger-auditing system, explicitly not a drive, a propulsion device, or a faster-travel claim. The closure package that would change the verdict is written out in Appendix A — invariant distance, a matched source ledger, an energy-condition audit with no hidden

exoticity, gauge and causality and backreaction passes, the UV assumption resolved, the aggregation barrier cleared, and an independent hostile reproduction. Until every one of those closes, the honest label is the one this note carries: a local metric-shortening toy profile, defined and screened — not a Mars distance reduced by a thousand.

Appendix A — The toy, the two routes, and where the engine kills it

This appendix writes out the toy metric, the two routes that meet on its geometry, the forced source that the geometry demands, the point at which the worked candidate dies, and the closure package that would be needed to change the verdict. The geometry is undergraduate; the discipline is the work.

A.1 The metric and the ruler-density

Proper length is what the metric says a ruler measures; the proper-length density is the square root of the metric component:

$$d\ell = \sqrt{g_{xx}} dx, \quad \rho_L(x) \equiv \sqrt{g_{xx}(x)} \quad (\text{proper-length density})$$

Setting it small in the forward lobe is the whole toy — a thousand coordinate kilometres measuring one kilometre of ruler-length:

$$g_{xx} = 10^{-6} \Rightarrow \rho_L = 10^{-3} \Rightarrow 1000 \text{ km coordinate} \longrightarrow 1 \text{ km proper}$$

A.2 The proper length and the local shortening score

Integrating the density over an interval gives the proper length, and the local shortening score compares the compressed lobe to ordinary space — a 99.9% local deficit for the toy:

$$L = \int_a^b \sqrt{g_{xx}} dx, \quad S_L = 1 - \frac{L_{\text{compressed}}}{L_{\text{normal}}} = 1 - 10^{-3} = 0.999 \quad (\text{local, toy})$$

A.3 The front-lobe profile

A smooth, vehicle-comoving compression profile dips toward 10^{-3} inside the lobe and returns to 1 outside:

$$\lambda(\xi) = 1 - (1 - 10^{-3}) e^{-(\xi - \xi_0)^2 / 2\sigma^2}, \quad g_{\xi\xi} = \lambda(\xi)^2, \quad \xi = x - x_{\text{ship}}(t)$$

A.4 The two routes meet on the geometry

The 13D route projects the candidate profile into the ordinary 4D metric and evaluates the same proper-distance functional — so it agrees with the ordinary-GR route on the local geometry:

$$\Xi_{\text{drive}} \xrightarrow{\text{project}} g_{\mu\nu}^{4D} = g_{\mu\nu}^{\text{bg}} + \delta g_{\mu\nu}, \quad D[\Xi] = \int_{\text{front}} \sqrt{g_{\xi\xi}} d\xi = \int_{\text{front}} \lambda(\xi) d\xi$$

A.5 The forced source — and where MID-METRIC-001 dies

Einstein’s equation, read backward, makes the source the geometry demands non-optional:

$$G_{\mu\nu} = 8\pi G T_{\mu\nu} \implies T_{\mu\nu}^{\text{req}} = \frac{1}{8\pi G} G_{\mu\nu}[g] \quad (\text{the source is not optional})$$

For the worked candidate, that demanded source is exotic — a negative effective energy density violating the null, weak, and dominant energy conditions, supplied by no declared field. The source gate fails:

$$G_{tt} = -2.0202 \times 10^{-4}, \quad \rho_{\text{eff}} = -8.038 \times 10^{-6} < 0 \quad (\text{NEC/WEC/DEC violated}) \Rightarrow \text{FAIL_UNSOURCED_METRIC}$$

A.6 The no-global-aggregation barrier

Even a real local shortening does not integrate into a global travel-time cut; building a Mars time directly from the local score is blocked:

$$\underbrace{S_L}_{\text{local shortening}} \not\Rightarrow \underbrace{\Delta t_{\text{Mars}}}_{\text{global travel time}} \Rightarrow \text{FAIL_MISSION_OVERCLAIM}$$

A.7 The closure package, and what rests where

The local metric target is mathematically simple; the hard problem is paying for it. The package that would be needed to change the verdict is:

- an invariant-distance pass and a matched source ledger (a declared field that supplies the demanded stress-energy);
- an energy-condition audit with no hidden exoticity, or an explicitly allowed source; gauge / pure-coordinate artifact rejection;
- causality / horizon and EFT / backreaction passes; UV / moduli assumptions resolved or honestly delegated;
- the no-global-aggregation barrier addressed without overclaim; and an independent, hostile reproduction.
- **What this note shows.** The toy is easy to write, and the two routes compute the same local deficit — a target-profile and screening result.
- **What the engine establishes.** On its own best worked candidate, the demanded source is exotic and unsourced, so the candidate is killed at the source gate; the ceiling never rises above a toy label, and every incompleteness can only downgrade a candidate.
- **What is out of scope by construction.** A working drive, a hardware or energy-source recipe, faster-than-light travel, and any Mars travel-time claim — the closure ladder’s physical-claim rung is targeted nowhere.

Screened result. The 1000× front-lobe toy is a well-defined, idealized metric profile whose local proper-distance deficit the ordinary-GR and 13D routes compute identically — and which, like the corpus’s honestly worked candidate MID-METRIC-001, *fails the source and energy gates* (exotic, unsourced stress-energy) and is blocked by the no-global-aggregation barrier from any travel-time claim. Verdict: MARS_FRONT_LOBE_1000X_TOY_PROFILE_DEFINED_AND_SCREENED — a screening result at the PASS_TOY ceiling, not a drive, not a Mars distance reduced 1000×, and not a physical claim. The value is the engine’s discipline: it kills its own best candidate rather than overclaim.

Notes and sources

[1] The First-Failure Engine, the ten-gate fail-closed screening pipeline, the worked exemplar MID-METRIC-001 (a real $\sim 0.22\%$ invariant local deficit with $R \neq 0$, killed by $\rho_{\text{eff}} = -8.038 \times 10^{-6}$ with NEC/WEC/DEC violated, terminal token FAIL_UNSOURCED_METRIC), the PASS_TOY ceiling, and the no-global-aggregation result are from the attached Mars metric-interface design package and the provisional patent. The platform is a screening / source-ledger-auditing system, explicitly not a drive, propulsion device, exotic-matter source, or faster-travel claim.

[2] Provisional patent: Bergstrom, C., U.S. Provisional Patent Application, ‘System and Method for Native-Geometry (Thirteen-Dimensional) Admissibility Screening and Fail-Closed Energy-Ledger Auditing of Candidate Spacetime-Metric Engineering Profiles’ (attached; correspondence via physics.magflowmeters.com). Governing scope: a computer-implemented screening, classification, and source-ledger-auditing platform whose ceiling for any candidate is a status-labeled simulation result (PASS_TOY); it is structurally incapable of certifying physical realizability, and that incapability is the intended, load-bearing property.

[3] The proper-length functional $L = \int \sqrt{g_{xx}} dx$, the forced-source identity $T_{\text{req}} = G[g]/8\pi G$, and the classical energy conditions (NEC/WEC/SEC/DEC) are standard general relativity; see any GR text.

[4] The 1000× toy in this note is a deliberately idealized, larger illustration; the corpus’s honestly worked candidate (MID-METRIC-001) is the smaller $\sim 0.22\%$ deficit, and it fails the source and energy gates. Neither is a physical claim; the closure ladder’s physical-claim rung is out of scope everywhere.

[5] Companion notes: the five force and quantum-gravity consistency notes, the synthesis (*One Geometry, Four Forces*), the quantum-computing noise check (*Bit-Flip Code*), and the particle-ontology note (*Why Only These Particles?*), same series.